

FERNANDO PALERO MOLINA

fpalero1986@gmail.com | +49 176 81268878 | Alemania | LinkedIn: [://linkedin.com](https://www.linkedin.com/in/fpalero1986)

PROFILE HEADLINE

Senior Software Engineer | Java & Microservices Specialist | Cloud Architect | AI & Multi-Agent Systems Engineer | Technical Team Lead

PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer and Technical Team Lead with over 12 years of experience designing, scaling, and delivering high-performance backend systems and advanced AI/LLM applications. Expert in Java microservices, cloud architectures (AWS & Azure), and software craftsmanship (DDD, Hexagonal Architecture, SOLID). Pioneer in building complex full-stack multi-agent AI solutions using Python, LangChain, LangGraph, LlamaIndex, Angular, and Prisma. Proven track record leading monolith-to-microservices migrations, optimizing cloud infrastructure, and mentoring high-performing engineering teams in international remote environments.

SKILLS

AI & Intelligent Systems: Multi-Agent Architectures, LangGraph, LangChain, LlamaIndex, Prompt Engineering, Local LLMs, Retrieval-Augmented Generation (RAG), Machine Learning Metrics.

Programming & Architecture: Java, Python, Spring Boot, Spring Cloud, Microservices, Modular Architecture, SOLID, DDD, Hexagonal Architecture, REST APIs, Kafka, Prisma.

Databases & Cloud: PostgreSQL, MongoDB, AWS (ECS, EC2, S3), Azure (Event Hub, Cosmos DB, Functions, Azure Authorization), Nextcloud, Appsmith.

Frontend, Mobile & Tools: Angular, React, EmberJS, TypeScript, JavaScript, Android SDK, Docker, Docker Compose, Devcontainer, GitHub Actions, GitFlow, CI/CD.

Methodologies & Testing: Agile, Scrum, Test-Driven Development (TDD), Automation Testing, PyTest, Jasmine, Karma, Selenium, JUnit, Mockito, Cypress.

Leadership: Technical Mentorship, Cross-Functional Leadership, Architecture Governance, Code Reviews.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Material One | Remote, Germany

July 2025 - December 2025

- Led the greenfield development of a high-throughput data aggregation platform, establishing core Java architecture patterns and engineering standards for multi-source data integration.
- Designed and built scalable backend microservices using Java, Spring Boot, and Spring Cloud, leveraging Domain-Driven Design (DDD) to achieve 99.9% system availability.
- Engineered resilient cloud infrastructure using Microsoft Azure, integrating Event Hub for real-time Java streaming components, Cosmos DB, and serverless Azure Functions with secure Azure Authorization.
- Built responsive full-stack capabilities with Angular and TypeScript, reducing data loading times on monitoring dashboards by 25%.
- Containerized Java development environments using Docker, Docker Compose, and Devcontainers, cutting team onboarding time by 40%.
- Developed high-performance event-driven schemas using Kafka and optimized relational databases using PostgreSQL.
- Attained 90%+ code coverage by implementing robust automated testing pipelines with JUnit and Mockito, reducing production bugs by 15%.

Software Engineer Team Lead | Aevi | Prague, Czech Republic (Remote)

February 2022 - February 2025

- Spearheaded the technical migration from a legacy monolithic architecture to a modern Java microservices and modular ecosystem, reducing infrastructure operational costs by 30%.
- Directed and mentored an international team of 5 engineers, improving overall sprint delivery velocity by 20% through agile coaching and rigorous Java code reviews.
- Architected resilient, cloud-native microservices based on Java, Hexagonal Architecture, and DDD principles to ensure decoupled domain logic.
- Deployed, managed, and scaled containerized enterprise Java services across AWS cloud infrastructure using ECS and EC2.
- Modernized core frontend modules using React and TypeScript, improving overall application responsiveness and user satisfaction scores.
- Established CI/CD automation pipelines via GitHub Actions, decreasing software deployment cycles from days to minutes.
- Built automated end-to-end and integration testing matrices utilizing Java, JUnit, Mockito, and Cypress, eliminating manual

regression testing bottlenecks.

Software Engineer Team Lead | Saitow AG (Alzura) | Spain

September 2019 - January 2022

- Designed and implemented highly scalable microservices-based architectures using Java for a high-traffic B2B e-commerce platform processing thousands of transactions daily.
- Led end-to-end feature delivery, maximizing cloud resource utilization and improving backend system responsiveness by 15%.
- Enforced clean coding standards, SOLID principles, and architecture governance, significantly reducing technical debt.
- Successfully migrated services to AWS (ECS, EC2) using Docker and Docker Compose, introducing Devcontainers to standardize environment parity across teams.
- Supervised full-stack feature alignments using React and TypeScript to ensure seamless frontend-backend integration.
- Secured continuous delivery guardrails through GitHub Actions and drove rigorous data storage optimizations using PostgreSQL.
- Guaranteed application robustness through extensive automated test suites using JUnit, Mockito, and Cypress.

Software Analyst | Backend Developer | Koukio | Spain

April 2017 - August 2019

- Enhanced core banking software (ABN AMRO), ensuring high availability, bulletproof microservices security, and 100% compliance with strict financial regulations.
- Developed scalable enterprise features using Java, JBoss, Spring Boot, and Spring Cloud.
- Transitioned on-premise financial workflows to AWS (ECS, EC2), operating under Domain-Driven Design (DDD) to support concurrent transactions.
- Standardized containerized environments using Docker, Docker Compose, and Devcontainers to guarantee development consistency.
- Maintained full-stack capabilities by engineering user interface components with Angular and TypeScript.
- Automated software quality guardrails through high-coverage testing frameworks using JUnit, Mockito, and Cypress.
- Integrated high-performance data storage systems handling robust relational schemas with PostgreSQL.
- Streamlined continuous integration workflows by creating automated CI/CD pipelines via GitHub Actions.
- Built a predictive AI-based error forecasting module using Python and TensorFlow, proactively preventing critical system downtime.

Software Analyst | Backend Developer | Sopra Steria | Spain

March 2016 - March 2017

- Developed and optimized mobile applications for FERMAX intercom hardware solutions utilizing Java and Android SDK, improving performance on legacy devices.
- Engineered responsive frontend web layouts for the E.Leclerc retail platform utilizing EmberJS and JavaScript.
- Created end-to-end automation frameworks using Selenium, JUnit, and Mockito, accelerating QA delivery release cycles by 25%.
- Refactored tightly coupled legacy modules using SOLID design principles, improving system maintainability and readability.

Project Manager | Telefónica | Spain

January 2011 - January 2012

- Coordinated cross-functional engineering teams to implement infrastructure upgrades, telecom system optimizations, and platform modernization initiatives ahead of schedule.

TECHNICAL PROJECTS

Lead AI Engineer & Full-Stack Developer | Open-Source Intelligent Applications | Personal Portfolio

January 2026 - Present

- Automated Order Management Chatbot: Conceptualized and built an autonomous conversational agent utilizing Python, LangChain, and local LLMs to handle commercial orders. Developed the frontend layout with Angular and TypeScript, leveraging Appsmith for rapid internal tool orchestration and Nextcloud for secure cloud file management. Integrated a robust data layer with Prisma managing a PostgreSQL database. Containerized the entire infrastructure using Docker, Docker Compose, and Devcontainers, automating cloud deployment to AWS EC2 instances. Guaranteed system stability using Python testing libraries (PyTest/Unittest) and Angular testing frameworks (Jasmine/Karma) to achieve full pipeline automation.
- Multi-Agent Personal Coach & Trainer Platform: Engineered a sophisticated multi-agent system using LangGraph, LangChain, and LlamaIndex for semantic RAG data retrieval. Built the full-stack user interface with Angular, TypeScript, and Appsmith, utilizing Nextcloud and AWS S3 to store and sync raw multimedia athlete data. Designed a dual-database architecture using Prisma with PostgreSQL for transactional user metrics and MongoDB for unstructured fitness data analytics. Standardized the development workflow via Devcontainers and Docker Compose pipelines, ensuring cross-environment application reliability through comprehensive automated testing suites across Python and Angular.

- IPSF Sports Advisor: Designed a specialized regulatory compliance assistant for international pole sports athletes, powered by Claude and running on a multi-agent framework with LangGraph, LangChain, and LlamaIndex. Architected the full-stack application using Angular and TypeScript for the user interface, Appsmith for dashboard visualization, and Nextcloud for document federation. Managed rulebook validation structures using Prisma connected to MongoDB and PostgreSQL instances. Deployed the containerized application on scalable AWS container services using Docker, validating system behavior through automated Python and Angular test matrices.

LANGUAGES

English: Full Professional Proficiency
Spanish: Native / Bilingual Proficiency
German: Beginner/Basic Proficiency

EDUCATION

Master of Science (M.Sc.) - ICT Research | Autonomous University of Madrid, Spain
Bachelor of Science (B.Sc.) - Computer Science Engineering | Polytechnic University of Valencia, Spain

CERTIFICATIONS

Scrum Master